67 Disorders of Granulocytes and Monocytes

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Leukocytes, the major cells comprising inflammatory and immune responses, include neutrophils, T and B lymphocytes, natural killer (NK) cells, mononuclear phagocytes (blood monocytes and tissue macrophages), eosinophils, and basophils. These cells have specific functions, such as antibody production by B lymphocytes or destruction of bacteria by neutrophils, but in no single infectious disease is the exact role of the cell types completely established. Thus, whereas neutrophils are classically thought to be critical to host defense against bacteria, they may also play important roles in defense against viral infections.

The blood delivers leukocytes to the various tissues from the bone marrow, where they are produced. Normal blood leukocyte counts are 4.3–10.8 × 10^9/L, with neutrophils representing 45–74% of the cells, bands 0–4%, lymphocytes 16–45%, monocytes 4–10%, eosinophils 0–7%, and basophils 0–2%. Variation among individuals and among different ethnic groups can be substantial, with lower leukocyte numbers for certain African-American ethnic groups. Lower granulocyte numbers in African Americans are often in the 1500–2000/μL range and are generally without sequelae, a condition termed benign ethnic neutropenia. The lower number of granulocytes is associated with null expression of the Duffy antigen receptor for cytokines (*DARC*) gene, a receptor for *Plasmodium vivax*, the absence of which conveys resistance to this form of malaria. The various leukocytes are derived from a common stem cell in the bone marrow. Three-fourths of the nucleated cells of bone marrow are committed to the production of leukocytes. Leukocyte maturation in the marrow is under the regulatory control of a number of different factors, known as colony-stimulating factors (CSFs) and interleukins (ILs). Because an alteration in the number and type of leukocytes is often associated with disease processes, total white blood cell (WBC) count (cells per μL) and differential counts are informative. This chapter focuses on neutrophils, monocytes, and eosinophils. **Lymphocytes and basophils are discussed in Chaps. 360 and 364, respectively.**

NEUTROPHILS

MATURATION

Important events in neutrophil life are summarized in **Fig. 67-1**. In normal humans, neutrophils are produced only in the bone marrow. The minimum number of stem cells necessary to support hematopoiesis is estimated to be 400–500 at any one time. Human blood monocytes, tissue macrophages, and stromal cells produce CSFs, hormones required for the growth of monocytes and neutrophils in the bone marrow. The hematopoietic system not only produces enough neutrophils (~1.3 × 10^{11} cells per 80-kg person per day) to carry out physiologic functions but also has a large reserve stored in the marrow, which can be mobilized in response to inflammation or infection. An increase in the number of blood neutrophils is called *neutrophilia*, and the presence of immature cells is termed a *shift to the left*. A decrease in the number of blood neutrophils is called *neutropenia*.

Neutrophils and monocytes evolve from pluripotent stem cells under the influence of cytokines and CSFs **(Fig. 67-2)**. The proliferation phase through the metamyelocyte takes about 1 week, while the maturation phase from metamyelocyte to mature neutrophil takes another week. The myeloblast is the first recognizable precursor cell and is followed by the *promyelocyte*. The promyelocyte evolves when the classic lysosomal granules, called the *primary*, or *azurophil, granules* are produced. The primary granules contain hydrolases, elastase, myeloperoxidase, cathepsin G, cationic proteins, and bactericidal/permeability-increasing protein, which is important for killing gram-negative bacteria. Azurophil granules also contain *defensins*, a family of cysteine-rich polypeptides with broad antimicrobial activity against bacteria, fungi, and certain enveloped viruses. The promyelocyte divides to produce the *myelocyte*, a cell responsible for the synthesis of the *specific*, or *secondary, granules*, which contain unique (specific) constituents such as lactoferrin, vitamin B_{12}–binding protein, membrane components of the reduced nicotinamide-adenine dinucleotide phosphate (NADPH) oxidase (NOX2) required for hydrogen peroxide production, histaminase, and receptors for certain chemoattractants and adherence-promoting factors (e.g., CR3) as well as receptors for the basement membrane component, laminin. The secondary granules do not contain acid hydrolases and therefore are not classic lysosomes. Packaging of secondary granule contents during myelopoiesis is controlled by CCAAT/enhancer binding protein-ε (encoded by *CEBPE*). Secondary granule contents are readily released extracellularly, and their mobilization is important in modulating inflammation. During the final stages of maturation, no cell division occurs, and the cell passes through the metamyelocyte stage and then to the band neutrophil with a sausage-shaped nucleus **(Fig. 67-3)**. As the band cell matures, the nucleus assumes a lobulated configuration. The nucleus of neutrophils normally contains up to four segments **(Fig. 67-4)**. Excessive segmentation (>5 nuclear lobes) may be a manifestation of folate or vitamin B_{12} deficiency or the congenital neutropenia syndrome of warts, hypogammaglobulinemia, infections, and myelokathexis (WHIM) described below. The Pelger-Hüet anomaly **(Fig. 67-5)**, an infrequent dominant benign inherited trait caused by heterozygous mutations in the lamin B receptor, results in neutrophils with distinctive bilobed nuclei that must be distinguished from band forms. Acquired bilobed nuclei, pseudo-Pelger-Hüet anomaly, can occur with acute infections or in myelodysplastic syndromes. The physiologic role of the normal multilobed nucleus of neutrophils is unknown, but it may allow great deformation of neutrophils during migration into tissues at sites of inflammation and facilitate production of neutrophil nets.

In severe acute bacterial infection, prominent neutrophil cytoplasmic granules, called *toxic granulations*, are occasionally seen. Toxic granulations are immature or abnormally staining azurophil granules. Cytoplasmic inclusions, also called *Döhle bodies* (Fig. 67-3), can be seen during infection and are fragments of ribosome-rich endoplasmic reticulum. Large neutrophil vacuoles are often present in acute bacterial infection in some viral infections such as COVID-19 and probably represent pinocytosed (internalized) membrane **(Fig. 67-6)**.

Neutrophils are heterogeneous in function. Monoclonal antibodies have been developed that recognize only a subset of mature neutrophils. The meaning of neutrophil heterogeneity is not known.

The morphology of eosinophils is shown in **Fig. 67-7**.

MARROW RELEASE AND CIRCULATING COMPARTMENTS

Specific signals, including IL-1, tumor necrosis factor α (TNF-α), the CSFs, complement fragments, and chemokines, mobilize leukocytes from the bone marrow and deliver them to the blood in an unstimulated state. Under normal conditions, ~90% of the neutrophil pool is in the bone marrow, 2–3% in the circulation, and the remainder in the tissues **(Fig. 67-8)**.

The circulating pool exists in two dynamic compartments: one freely flowing and one marginated. The freely flowing pool is about one-half the neutrophils in the basal state and is composed of those cells that are in the blood and not in contact with the endothelium. Marginated leukocytes are those that are in close physical contact with the endothelium **(Fig. 67-9)**. In the pulmonary circulation, where an extensive capillary bed (~1000 capillaries per alveolus) exists, margination occurs because the capillaries are about the same size as a mature neutrophil. Therefore, neutrophil fluidity and deformability are necessary to make the transit through the pulmonary bed. Increased neutrophil rigidity and decreased deformability lead to augmented neutrophil trapping and margination in the lung. In contrast, in the systemic postcapillary venules, margination is mediated by the interaction of specific

*Deceased

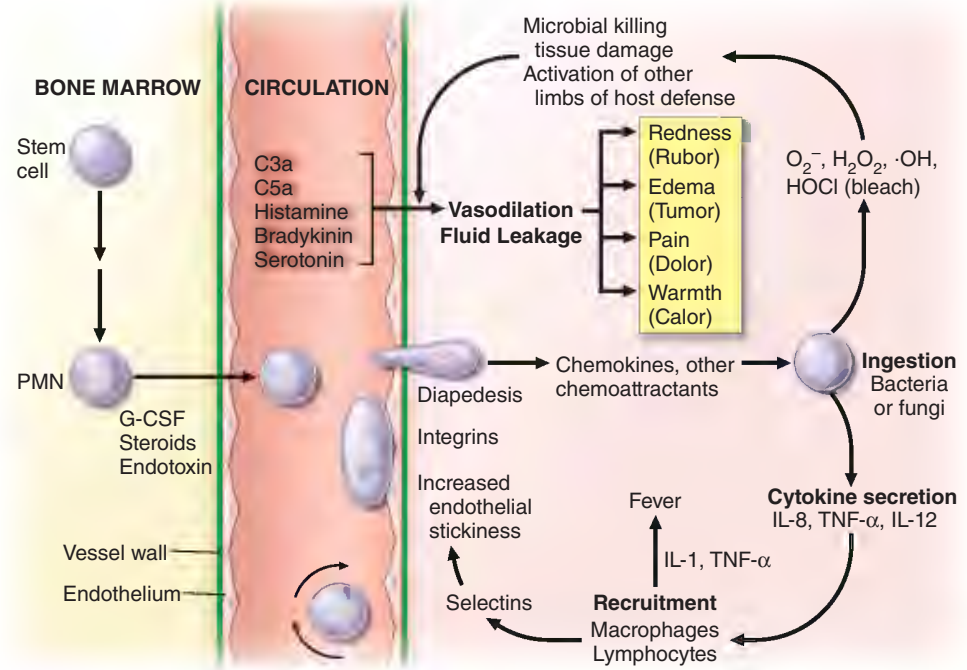
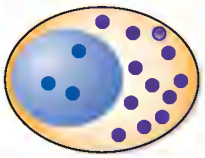
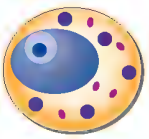
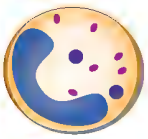
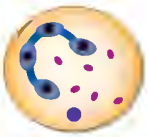


FIGURE 67-1 Schematic events in neutrophil production, recruitment, and inflammation. The four cardinal signs of inflammation (rubor, tumor, calor, dolor) are indicated, as are the interactions of neutrophils with other cells and cytokines. G-CSF, granulocyte colony-stimulating factor; IL, interleukin; PMN, polymorphonuclear leukocyte; TNF-α, tumor necrosis factor α.

Cell	Stage	Surface Markers[a]	Characteristics
	MYELOBLAST	CD33, CD13, CD15	Prominent nucleoli
	PROMYELOCYTE	CD33, CD13, CD15	Large cell Primary granules appear
	MYELOCYTE	CD33, CD13, CD15, CD14, CD11b	Secondary granules appear
	METAMYELOCYTE	CD33, CD13, CD15, CD14, CD11b	Kidney bean–shaped nucleus
	BAND FORM	CD33, CD13, CD15, CD14, CD11b, CD10, CD16	Condensed, band–shaped nucleus
	NEUTROPHIL	CD33, CD13, CD15, CD14, CD11b, CD10, CD16	Condensed, multilobed nucleus

[a]CD = Cluster Determinant; ● Nucleolus; ● Primary granule; ● Secondary granule.

FIGURE 67-2 Stages of neutrophil development shown schematically. Granulocyte colony-stimulating factor (G-CSF) and granulocyte-macrophage colony-stimulating factor (GM-CSF) are critical to this process. Identifying cellular characteristics and specific cell-surface markers are listed for each maturational stage.

cell-surface molecules called *selectins*. Selectins are glycoproteins expressed on neutrophils and endothelial cells, among others, that cause a low-affinity interaction, resulting in "rolling" of the neutrophil along the endothelial surface. On neutrophils, the molecule L-selectin (cluster determinant [CD] 62L) binds to glycosylated proteins on endothelial cells (e.g., glycosylation-dependent cell adhesion molecule [GlyCAM1] and CD34). Glycoproteins on neutrophils, most importantly sialyl-Lewis^x (SLe^x, CD15s), are targets for binding of selectins expressed on endothelial cells (E-selectin [CD62E] and P-selectin [CD62P]) and other leukocytes. In response to chemotactic stimuli from injured tissues (e.g., complement product C5a, leukotriene B_4, IL-8) or bacterial products (e.g., *N*-formylmethionylleucylphenylalanine [f-met-leu-phe of fMLF]), neutrophil adhesiveness increases through mobilization of intracellular adhesion proteins stored in specific granules to the cell surface, and the cells "stick" to the endothelium through *integrins*. The integrins are leukocyte glycoproteins that exist as complexes of a common CD18 β chain with CD11a (LFA-1), CD11b (called Mac-1, CR3, or the C3bi receptor), and CD11c (called p150,95 or CR4). CD11a/CD18 and CD11b/CD18 bind to specific endothelial receptors (intercellular adhesion molecules [ICAM] 1 and 2).

On cell stimulation, L-selectin is shed from neutrophils, and E-selectin increases in the blood, presumably because it is shed from endothelial cells; receptors for chemoattractants and opsonins are mobilized; and the phagocytes orient toward the chemoattractant source in the extravascular space, increase their motile activity (chemokinesis), and migrate directionally (chemotaxis) into tissues. The process of migration into tissues is called *diapedesis* and involves the crawling of neutrophils between postcapillary endothelial cells that open junctions between adjacent cells to permit leukocyte passage. Diapedesis involves platelet/endothelial cell adhesion molecule (PECAM) 1 (CD31), which is expressed on both the emigrating leukocyte and the

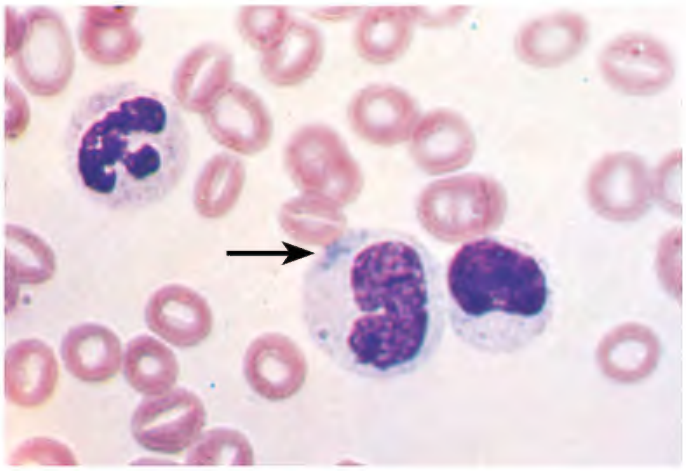


FIGURE 67-3 Neutrophil band with Döhle body. The neutrophil with a sausage-shaped nucleus in the center of the field is a band form. Döhle bodies (*arrow*) are discrete, blue-staining, nongranular areas found in the periphery of the cytoplasm of the neutrophil in infections and other toxic states. They represent aggregates of rough endoplasmic reticulum.

endothelial cells. The endothelial responses (increased blood flow from increased vasodilation and permeability) are mediated by anaphylatoxins (e.g., C3a and C5a) as well as vasodilators such as histamine, bradykinin, serotonin, nitric oxide, vascular endothelial growth factor (VEGF), and prostaglandins E and I. Cytokines regulate some of these processes (e.g., TNF-α induction of VEGF, interferon [IFN] γ inhibition of prostaglandin E).

In the healthy adult, most neutrophils leave the body by migration through the mucous membrane of the oral cavity and especially the gastrointestinal tract. Normally, neutrophils spend a short time in the circulation (half-life, 6–7 h). Senescent neutrophils are cleared from the circulation by macrophages in the lung and spleen. Once in the tissues, neutrophils release enzymes, such as collagenase and elastase, which may help establish abscess cavities. Neutrophils ingest pathogenic materials that have been opsonized by IgG and C3b. Fibronectin and the tetrapeptide tuftsin also facilitate phagocytosis.

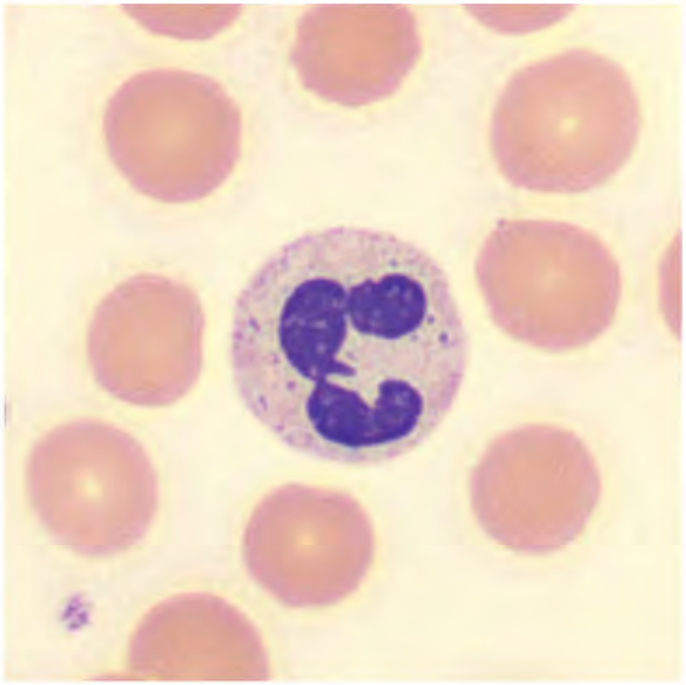


FIGURE 67-4 Normal granulocyte. The normal granulocyte has a segmented nucleus with heavy, clumped chromatin; fine neutrophilic granules are dispersed throughout the cytoplasm.

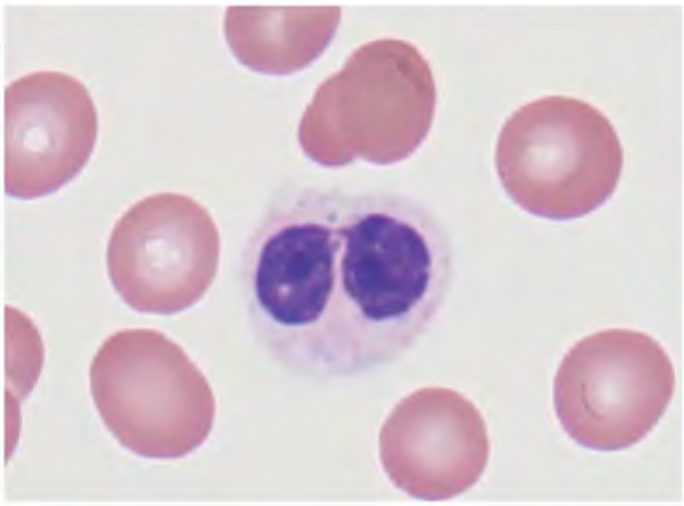


FIGURE 67-5 Pelger-Hüet anomaly. In this disorder, granulocytes are bilobed. In the benign, genetic form (due to mutations in *LBR*), most granulocytes have this appearance. In the acquired form, which is associated with myelodysplastic syndrome, it may be more occasional. The nucleus frequently has a spectacle-like, or "pince-nez," configuration. *(Courtesy of Dr. Katherine Calvo, Hematopathology Laboratory, Department of Laboratory Medicine, Clinical Center, NIH.)*

With phagocytosis comes a burst of oxygen consumption and activation of the hexose-monophosphate shunt. A membrane-associated NADPH oxidase called NOX2, consisting of membrane and cytosolic components, is assembled and catalyzes the univalent reduction of oxygen to superoxide anion, which is then converted by superoxide dismutase to hydrogen peroxide and other toxic oxygen products (e.g., hydroxyl radical). Hydrogen peroxide + chloride + neutrophil myeloperoxidase generates hypochlorous acid (bleach), hypochlorite, and chlorine. These products oxidize and halogenate microorganisms and tumor cells and, when uncontrolled, can damage host tissue. Strongly cationic proteins, defensins, elastase, cathepsins, and probably nitric oxide also participate in microbial killing. Lactoferrin chelates iron, an important growth factor for microorganisms, especially fungi. Other enzymes, such as lysozyme and acid proteases, help digest microbial debris. After 1–4 days in tissues, neutrophils die. The apoptosis of neutrophils is also cytokine-regulated; granulocyte colony-stimulating factor (G-CSF) and IFN-γ prolong their life span. Neutrophil extracellular traps (NETs) consisting of a DNA scaffold decorated with neutrophil-granule derived proteins, such as enzymatically active proteases and antimicrobial peptides, are thought to be formed as a defense mechanism to immobilize invading microorganisms. Under certain conditions, such as in delayed-type hypersensitivity, monocyte accumulation occurs within 6–12 h of initiation of inflammation. Neutrophils, monocytes, microorganisms in various states of digestion, and altered local tissue cells make up the inflammatory exudate, pus. Myeloperoxidase confers the characteristic green color to pus and may participate in turning off the inflammatory process by inactivating chemoattractants and immobilizing phagocytic cells.

Neutrophils respond to certain cytokines (IFN-γ, granulocyte-macrophage colony-stimulating factor [GM-CSF], IL-8) and produce cytokines and chemotactic signals (TNF-α, IL-8, macrophage inflammatory protein [MIP] 1) that modulate the inflammatory response. In the presence of fibrinogen, fMLF, or leukotriene B_4, IL-8 production by neutrophils is induced, providing autocrine amplification of inflammation. *Chemokines* (*chemo*attractant cyto*kines*) are small proteins produced by many different cell types, including endothelial cells, fibroblasts, epithelial cells, neutrophils, and monocytes, that regulate neutrophil, monocyte, eosinophil, and lymphocyte recruitment and activation. Chemokines transduce their signals through heterotrimeric G protein–linked receptors that have seven cell membrane–spanning domains, the same type of cell-surface receptor that mediates the response to the classic chemoattractants fMLF and C5a. Four major groups of chemokines are recognized based on the cysteine structure near the N terminus: C, CC, CXC, and CXXXC. The C chemokine

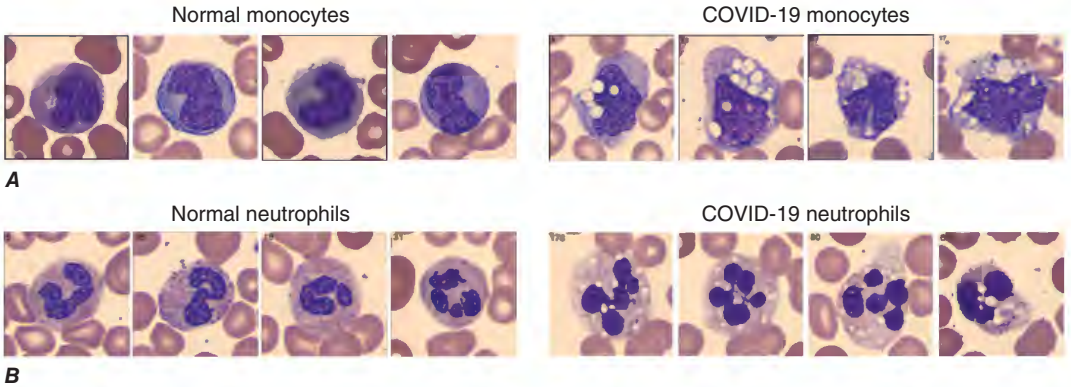


FIGURE 67-6 COVID-19: Vacuolization in peripheral blood monocytes and neutrophils of COVID-19 patients. Peripheral blood smear showing vacuolization in (***A***) monocytes and (***B***) neutrophils from hospitalized hypoxemic COVID-19 patients relative to healthy volunteers. Increased vacuoles were noted in ~80% of monocytes and ~50% of neutrophils in each COVID-19 patient throughout their hospitalization.

lymphotactin is T-cell tropic; CC chemokines such as MIP-1 attract lymphocytes, monocytes, eosinophils, and basophils; CXC cytokines such as IL-8 mainly attract neutrophils; while the CXXXC chemokine fractalkine attracts neutrophils, monocytes, and T cells. Not only do these molecules and their receptors regulate the trafficking and activation of inflammatory cells, but also specific chemokine receptors serve as co-receptors for HIV infection (Chap. 208), while others have roles in other viral infections (e.g., West Nile virus), susceptibility and response to *Candida*, and atherogenesis.

NEUTROPHIL ABNORMALITIES

Defects in the neutrophil life cycle can lead to dysfunction and compromised host defenses. When inflammation is severely depressed, the clinical result is often recurrent, severe bacterial and fungal infections. Aphthous ulcers of mucous membranes (gray ulcers without pus) and gingivitis and periodontal disease suggest a phagocytic cell disorder. Patients with congenital phagocyte defects can have infections within the first few days of life. Skin, ear, upper and lower respiratory tract, and bone infections are common. Sepsis and meningitis are rare. In some disorders, the frequency of infection is variable, and patients can go for months or even years without major infection. Aggressive management of these congenital diseases, including hematopoietic stem cell transplantation and gene therapy, has extended the life span of patients well into adulthood.

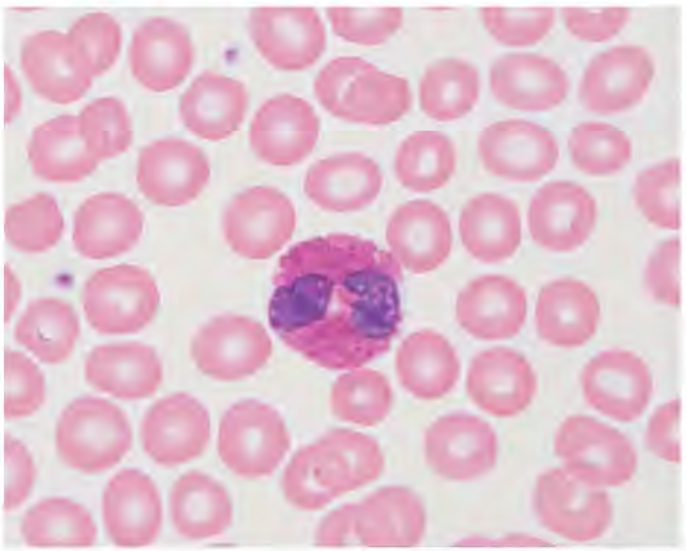


FIGURE 67-7 The normal eosinophil seen here contains large, bright orange granules and usually a bilobed nucleus. *(Courtesy of Dr. Katherine Calvo, Hematopathology Laboratory, Department of Laboratory Medicine, Clinical Center, NIH.)*

Neutropenia The consequences of absent neutrophils are dramatic. Susceptibility to infectious diseases increases sharply when neutrophil counts fall to <1000 cells/μL. When the absolute neutrophil count (ANC; band forms and mature neutrophils combined) falls to <500 cells/μL, control of endogenous microbial flora (e.g., mouth, gut) is impaired; when the ANC is <200/μL, the local inflammatory process is absent. Neutropenia can be due to depressed production, increased peripheral destruction, or excessive peripheral pooling. A falling neutrophil count or a significant decrease in the number of neutrophils below steady-state levels, together with a failure to increase neutrophil counts in the setting of infection or other challenge, requires investigation. Acute neutropenia, such as that caused by cancer chemotherapy, is more likely to be associated with increased risk of infection than chronic neutropenia (months to years) that reverses in response to infection or carefully controlled administration of endotoxin (see "Laboratory Diagnosis and Management," below).

Some causes of inherited and acquired neutropenia are listed in Table 67-1. The most common neutropenias are iatrogenic, resulting from the use of cytotoxic or immunosuppressive therapies for malignancy or control of autoimmune disorders. These drugs cause

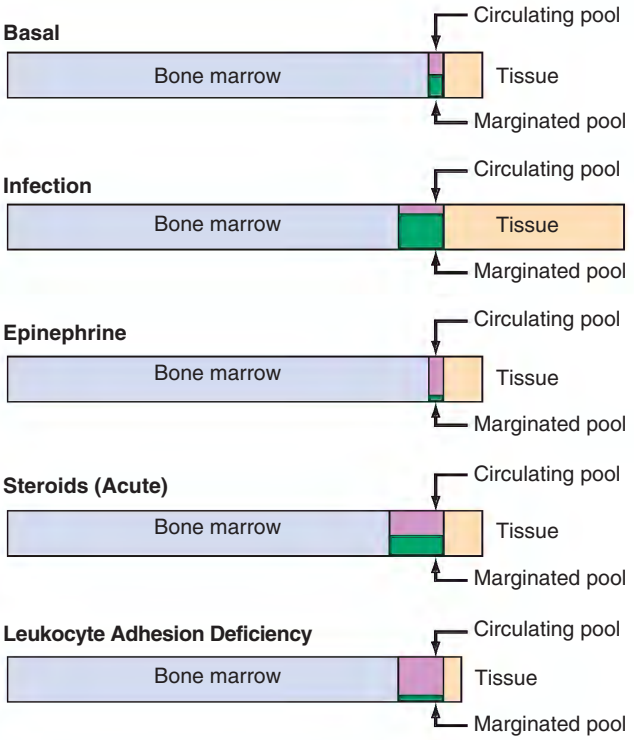


FIGURE 67-8 Schematic neutrophil distribution and kinetics between the different anatomic and functional pools.

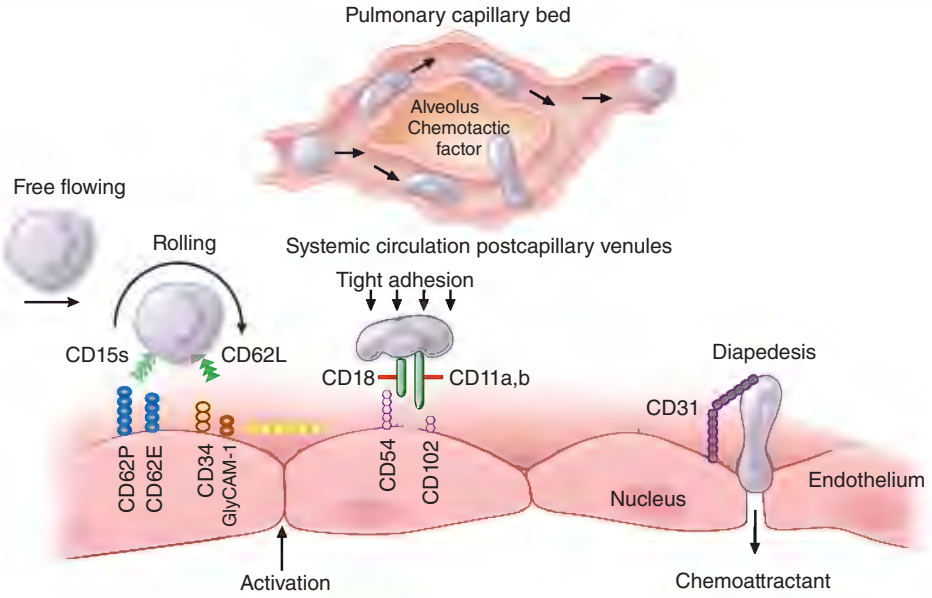


FIGURE 67-9 Neutrophil travel through the pulmonary capillaries is dependent on neutrophil deformability. Neutrophil rigidity (e.g., caused by C5a) enhances pulmonary trapping and response to pulmonary pathogens in a way that is not so dependent on cell-surface receptors. Intraalveolar chemotactic factors, such as those caused by certain bacteria (e.g., *Streptococcus pneumoniae*), lead to diapedesis of neutrophils from the pulmonary capillaries into the alveolar space. Neutrophil interaction with the endothelium of the systemic postcapillary venules is dependent on molecules of attachment. The neutrophil "rolls" along the endothelium using selectins: neutrophil CD15s (sialyl-Lewis^x) binds to CD62E (E-selectin) and CD62P (P-selectin) on endothelial cells; CD62L (L-selectin) on neutrophils binds to CD34 and other molecules (e.g., GlyCAM-1) expressed on endothelium. Chemokines or other activation factors stimulate integrin-mediated "tight adhesion": CD11a/CD18 (LFA-1) and CD11b/CD18 (Mac-1, CR3) bind to CD54 (ICAM-1) and CD102 (ICAM-2) on the endothelium. Diapedesis occurs between endothelial cells: CD31 (PECAM-1) expressed by the emigrating neutrophil interacts with CD31 expressed at the endothelial cell-cell junction. CD, cluster determinant; GlyCAM, glycosylation-dependent cell adhesion molecule; ICAM, intercellular adhesion molecule; PECAM, platelet/endothelial cell adhesion molecule.

neutropenia because they result in decreased production of rapidly growing progenitor (stem) cells of the marrow. Certain antibiotics such as chloramphenicol, trimethoprim-sulfamethoxazole, flucytosine, vidarabine, and the antiretroviral drug zidovudine may cause neutropenia by inhibiting proliferation of myeloid precursors. Azathioprine and 6-mercaptopurine are metabolized by the enzyme thiopurine methyltransferase (TMPT); hypofunctional polymorphisms that are found in 11% of whites can lead to accumulation of 6-thioguanine and profound marrow toxicity, which is why pharmacogenomic testing is recommended before initiating these drugs. The marrow suppression is generally dose-related and resolves after cessation of the drug.

Another important mechanism for iatrogenic neutropenia is the effect of drugs that serve as immune haptens and sensitize neutrophils or neutrophil precursors to immune-mediated peripheral destruction. This form of drug-induced neutropenia can be seen within 7 days of exposure to the drug; with previous drug exposure, resulting in preexisting antibodies, neutropenia may occur a few hours after administration of the drug. Although any drug can cause this form of neutropenia, the most frequent causes are commonly used antibiotics, such as sulfa-containing compounds, penicillins, and cephalosporins. Fever and eosinophilia may also be associated with drug reactions, but often these signs are not present. Drug-induced neutropenia can be severe, but discontinuation of the sensitizing drug is sufficient for recovery, which is usually seen within 5–7 days and is complete by 10 days. Readministration of the sensitizing drug should be avoided, because abrupt neutropenia will often result. For this reason, diagnostic challenge should be avoided.

Autoimmune neutropenias caused by circulating antineutrophil antibodies are another form of acquired neutropenia that results in increased destruction of neutrophils. Acquired neutropenia may also be seen with viral infections, including acute infection with HIV. Acquired neutropenia may be cyclic in nature, occurring at intervals of several weeks. Acquired cyclic or stable neutropenia may be associated with an expansion of large granular lymphocytes (LGLs), which may be T cells, NK cells, or NK-like cells. Patients with large granular lymphocytosis may have moderate blood and bone marrow lymphocytosis, neutropenia, polyclonal hypergammaglobulinemia, splenomegaly, rheumatoid arthritis, and absence of lymphadenopathy. Such patients may have a chronic and relatively stable course. Recurrent bacterial infections are frequent. Benign and malignant forms of this syndrome occur. In some patients, spontaneous regression has occurred even after 11 years, suggesting an immunoregulatory defect as the basis for at least one form of the disorder. Glucocorticoids, cyclosporine, methotrexate, and monoclonals are commonly used to manage these cytopenias.

TABLE 67-1 Causes of Neutropenia

Decreased Production
Drug-induced—alkylating agents (nitrogen mustard, busulfan, chlorambucil, cyclophosphamide); antimetabolites (methotrexate, 6-mercaptopurine, 5-flucytosine); noncytotoxic agents (antibiotics [chloramphenicol, penicillins, sulfonamides], phenothiazines, tranquilizers [meprobamate], anticonvulsants [carbamazepine], antipsychotics [clozapine], certain diuretics, anti-inflammatory agents, antithyroid drugs, many others)
Hematologic diseases—idiopathic, cyclic neutropenia, Chédiak-Higashi syndrome, aplastic anemia, infantile genetic disorders (see text)
Tumor invasion, myelofibrosis
Nutritional deficiency—vitamin B_{12}, folate (especially alcoholics)
Infection—tuberculosis, typhoid fever, brucellosis, tularemia, measles, infectious mononucleosis, malaria, viral hepatitis, leishmaniasis, AIDS
Peripheral Destruction
Antineutrophil antibodies and/or splenic or lung trapping
Autoimmune disorders—Felty syndrome, rheumatoid arthritis, lupus erythematosus
Drugs as haptens—aminopyrine, α-methyldopa, phenylbutazone, mercurial diuretics, some phenothiazines
Granulomatosis with polyangiitis (Wegener)
Peripheral Pooling (Transient Neutropenia)
Overwhelming bacterial infection (acute endotoxemia)
Hemodialysis
Cardiopulmonary bypass

Hereditary Neutropenias Hereditary neutropenias are rare and may manifest in early childhood as a profound constant neutropenia or agranulocytosis. Congenital forms of neutropenia include Kostmann's syndrome (neutrophil count <100/μL), which is often fatal and due to mutations in the anti-apoptosis gene *HAX-1*; severe chronic neutropenia (neutrophil count of 300–1500/μL) due to mutations in neutrophil elastase (*ELANE*); hereditary cyclic neutropenia or, more appropriately, cyclic hematopoiesis, also due to mutations in neutrophil elastase (*ELANE*); the cartilage-hair hypoplasia syndrome due to mutations in the mitochondrial RNA-processing endoribonuclease *RMRP*; Shwachman-Diamond syndrome associated with pancreatic insufficiency due to mutations in the Shwachman-Bodian-Diamond syndrome gene *SBDS*; the WHIM syndrome (*w*arts, *h*ypogammaglobulinemia, *i*nfections, *m*yelokathexis [retention of WBCs in the

marrow]), characterized by neutrophil hypersegmentation and bone marrow myeloid arrest due to mutations in the chemokine receptor *CXCR4*; and neutropenias associated with other immune defects, such as GATA2 deficiency, X-linked agammaglobulinemia, Wiskott-Aldrich syndrome, and CD40 ligand deficiency. Mutations in the G-CSF receptor can develop in severe congenital neutropenia and are linked to the development of leukemia. Absence of both myeloid and lymphoid cells is seen in reticular dysgenesis, due to mutations in the nuclear genome-encoded mitochondrial enzyme adenylate kinase-2 (*AK2*).

Maternal factors can be associated with neutropenia in the newborn. Transplacental transfer of IgG directed against antigens on fetal neutrophils can result in peripheral destruction. Drugs (e.g., thiazides) ingested during pregnancy can cause neutropenia in the newborn by either depressed production or peripheral destruction.

In Felty syndrome—the triad of rheumatoid arthritis, splenomegaly, and neutropenia **(Chap. 370)**—antibodies can shorten neutrophil life span, while large granular lymphocytes can attack marrow neutrophil precursors. Splenectomy may increase the neutrophil count in Felty syndrome and lower serum neutrophil-binding IgG. Some Felty syndrome patients also have autoantibodies to G-CSF, while others have increased numbers of LGLs. Splenomegaly with peripheral trapping and destruction of neutrophils is also seen in lysosomal storage diseases and commonly in portal hypertension.

Neutrophilia Neutrophilia results from increased neutrophil production, increased marrow release, or defective margination **(Table 67-2)**. The most important acute cause of neutrophilia is infection. Neutrophilia from acute infection represents both increased production and increased marrow release. Increased production is also associated with chronic inflammation and certain myeloproliferative diseases. Increased marrow release and mobilization of the marginated leukocyte pool are induced by glucocorticoids. Release of epinephrine, as with vigorous exercise, excitement, or stress, will demarginate neutrophils in the spleen and lungs and double the neutrophil count in minutes. Cigarette smoking can elevate neutrophil counts above the normal range. Leukocytosis with cell counts of 10,000–25,000/μL occurs in response to infection and other forms of acute inflammation and results from both release of the marginated pool and mobilization of marrow reserves. Persistent neutrophilia with cell counts of ≥30,000–50,000/μL is called a *leukemoid reaction*, a term often used to distinguish this degree of neutrophilia from leukemia. In a leukemoid reaction, the circulating neutrophils are usually mature and not clonally derived.

Abnormal Neutrophil Function Inherited and acquired abnormalities of phagocyte function are listed in **Table 67-3**. The resulting diseases are best considered in terms of the functional defects of adherence, chemotaxis, and microbicidal activity. The distinguishing features of the important inherited disorders of phagocyte function are shown in **Table 67-4**.

TABLE 67-2 Causes of Neutrophilia

Increased Production
Idiopathic
Drug-induced—glucocorticoids, G-CSF
Infection—bacterial, fungal, sometimes viral
Inflammation—thermal injury, tissue necrosis, myocardial and pulmonary infarction, hypersensitivity states, collagen vascular diseases
Myeloproliferative diseases—myelocytic leukemia, myeloid metaplasia, polycythemia vera
Increased Marrow Release
Glucocorticoids
Acute infection (endotoxin)
Inflammation—thermal injury
Decreased or Defective Margination
Drugs—epinephrine, glucocorticoids, nonsteroidal anti-inflammatory agents
Stress, excitement, vigorous exercise
Leukocyte adhesion deficiency type 1 (CD18); leukocyte adhesion deficiency type 2 (selectin ligand, CD15s); leukocyte adhesion deficiency type 3 (FERMT3)
Miscellaneous
Metabolic disorders—ketoacidosis, acute renal failure, eclampsia, acute poisoning
Drugs—lithium
Other—metastatic carcinoma, acute hemorrhage or hemolysis

Abbreviation: G-CSF, granulocyte colony-stimulating factor.

DISORDERS OF ADHESION Three main types of leukocyte adhesion deficiency (LAD) have been described. All are autosomal recessive and result in the impairment of neutrophil exit from the circulation to sites of infection, leading to leukocytosis and increased susceptibility to infection (Fig. 67-9). Patients with LAD 1 have mutations in *CD18*, the common component of the integrins LFA-1, Mac-1, and p150,95, leading to a defect in tight adhesion between neutrophils and the endothelium. The heterodimer formed by CD18/CD11b (Mac-1) is also the receptor for the complement-derived opsonin C3bi (CR3). The *CD18* gene is located on distal chromosome 21q. The severity of the defect determines the severity of clinical disease. Complete lack of expression of the leukocyte integrins results in a severe phenotype in which inflammatory stimuli do not increase the expression of leukocyte integrins on neutrophils or activated T and B cells. Neutrophils (and monocytes) from patients with LAD 1 adhere poorly to endothelial cells and protein-coated surfaces and exhibit defective spreading, aggregation, and chemotaxis. The inability of neutrophils to exit the vasculature to the tissue deprives the tissue macrophage of its expected neutrophil ingestion, leading to macrophage production of IL-23,

TABLE 67-3 Types of Granulocyte and Monocyte Disorders

	CAUSE OF INDICATED DYSFUNCTION		
FUNCTION	DRUG-INDUCED	ACQUIRED	INHERITED
Adherence-aggregation	Aspirin, colchicine, alcohol, glucocorticoids, ibuprofen, piroxicam	Neonatal state, hemodialysis	Leukocyte adhesion deficiency types 1, 2, and 3
Deformability		Leukemia, neonatal state, diabetes mellitus, immature neutrophils	
Chemokinesis-chemotaxis	Glucocorticoids (high dose), auranofin, colchicine (weak effect), phenylbutazone, naproxen, indomethacin, interleukin 2	Thermal injury, malignancy, malnutrition, periodontal disease, neonatal state, systemic lupus erythematosus, rheumatoid arthritis, diabetes mellitus, sepsis, influenza virus infection, herpes simplex virus infection, acrodermatitis enteropathica, AIDS	Chédiak-Higashi syndrome, neutrophil-specific granule deficiency, WDR1 deficiency, Job's syndrome (in some patients), Down syndrome, α-mannosidase deficiency, leukocyte adhesion deficiencies, Wiskott-Aldrich syndrome
Microbicidal activity	Colchicine, cyclophosphamide, glucocorticoids (high dose), TNF-α-blocking antibodies	Leukemia, aplastic anemia, certain neutropenias, tuftsin deficiency, thermal injury, sepsis, neonatal state, diabetes mellitus, malnutrition, AIDS	Chédiak-Higashi syndrome, neutrophil-specific granule deficiency, chronic granulomatous disease, defects in IFNγ/IL-12 axis, Anti-IFNγ autoantibodies

Abbreviations: IFNγ, interferon γ; IL, interleukin; TNF-α, tumor necrosis factor alpha.

TABLE 67-4 Inherited Disorders of Phagocyte Function: Differential Features

CLINICAL MANIFESTATIONS	CELLULAR OR MOLECULAR DEFECTS	DIAGNOSIS
Chronic Granulomatous Diseases (70% X-Linked, 30% Autosomal Recessive)		
Severe infections of skin, ears, lungs, liver, and bone with microorganisms such as *Staphylococcus aureus, Burkholderia cepacia* complex, *Aspergillus* spp., *Chromobacterium violaceum*; often hard to culture organism; excessive inflammation with granulomas, frequent lymph node suppuration; granulomas can obstruct GI or GU tracts; gingivitis, aphthous ulcers	No respiratory burst due to impaired of NADPH oxidase in neutrophils, monocytes, and eosinophils. Mutations in *CYBB, CYBA, NCF1, NCF2, NCF4,* or *CYBC1*	DHR or NBT test; no superoxide and H_2O_2 production by neutrophils; immunoblot for NADPH oxidase components; genetic detection
Chédiak-Higashi Syndrome (Autosomal Recessive)		
Recurrent pyogenic infections, especially with *S. aureus*; many patients get lymphoma-like illness during adolescence; periodontal disease; partial oculocutaneous albinism, nystagmus, progressive peripheral neuropathy, cognitive impairment in some patients	Reduced chemotaxis and phagolysosome fusion, increased respiratory burst activity, defective egress from marrow, abnormal skin window; defect in *CHS1*	Giant primary granules in neutrophils and other granule-bearing cells (Wright stain); genetic detection
Specific Granule Deficiency (Autosomal Recessive and Dominant)		
Recurrent infections of skin, ears, and sinopulmonary tract; delayed wound healing; decreased inflammation; bleeding diathesis	Abnormal chemotaxis, impaired respiratory burst and bacterial killing, failure to upregulate chemotactic and adhesion receptors with stimulation, defect in transcription of granule proteins. Mutations in *CEBPE* or *SMARCD2*	Lack of secondary (specific) granules in neutrophils (Wright stain), no neutrophil-specific granule contents (i.e., lactoferrin), no defensins, platelet α granule abnormality; genetic detection
Myeloperoxidase Deficiency (Autosomal Recessive)		
Clinically normal except in patients with underlying disease such as diabetes mellitus; then candidiasis or other fungal infections	No myeloperoxidase due to pre- and posttranslational defects in myeloperoxidase deficiency. Mutations in *MPO*	No peroxidase in neutrophils; genetic detection
Leukocyte Adhesion Deficiency		
Type 1: Delayed separation of umbilical cord, sustained neutrophilia, recurrent infections of skin and mucosa, gingivitis, periodontal disease	Impaired phagocyte adherence, aggregation, spreading, chemotaxis, phagocytosis of C3bi-coated particles; defective production of CD18 subunit common to leukocyte integrins. Mutations in *ITGB2*	Reduced phagocyte surface expression of the CD18-containing integrins with monoclonal antibodies against LFA-1 (CD18/CD11a), Mac-1 or CR3 (CD18/CD11b), p150,95 (CD18/CD11c); genetic detection
Type 2: Cognitive impairment, short stature, Bombay (hh) blood phenotype, recurrent infections, neutrophilia	Impaired phagocyte rolling along endothelium; due to defects in fucose transporter. Mutations in *SLC35C1*	Reduced phagocyte surface expression of Sialyl-Lewis^x, with monoclonal antibodies against CD15s; genetic detection
Type 3: Petechial hemorrhage, recurrent infections	Impaired signaling for integrin activation resulting in impaired adhesion. Mutations in *FERMT3*	Reduced signaling for adhesion through integrins; genetic detection
Phagocyte Activation Defects (X-Linked and Autosomal Recessive)		
NEMO deficiency: mild hypohidrotic ectodermal dysplasia; broad-based immune defect: pyogenic and encapsulated bacteria, viruses, *Pneumocystis*, mycobacteria; X-linked	Impaired phagocyte activation by IL-1, IL-18, TLR, CD40L, TNF-α leading to problems with inflammation and antibody production. Mutations in *IKBKG*	Poor in vitro response to endotoxin; impaired NF-κB activation; genetic detection
IRAK4 and MyD88 deficiency: susceptibility to pyogenic bacteria such as staphylococci, streptococci, clostridia; resistant to *Candida*; autosomal recessive	Impaired phagocyte activation by endotoxin through TLR and other pathways; TNF-α signaling preserved. Mutations in *IRAK4* or *MYD88*	Poor in vitro response to endotoxin; lack of NF-κB activation by endotoxin; genetic detection
Hyper IgE–Recurrent Infection Syndrome (Autosomal Dominant) (Job's Syndrome)		
Eczematoid or pruritic dermatitis, "cold" skin abscesses, recurrent pneumonias with *S. aureus* with bronchopleural fistulae and cyst formation, mild eosinophilia, mucocutaneous candidiasis, characteristic facies, restrictive lung disease, scoliosis, delayed primary dental deciduation	Reduced chemotaxis in some patients, reduced memory T and B cells. Mutations in *STAT3*	Somatic and immune features involving lungs, skeleton, and immune system; serum IgE >2000 IU/mL; genetic testing
DOCK8 deficiency (autosomal recessive), severe eczema, atopic dermatitis, cutaneous abscesses, HSV, HPV, and molluscum infections, severe allergies, cancer	Impaired T-cell proliferation to mitogens. Mutations in *DOCK8*	Severe allergies, viral infections, high IgE, eosinophilia, low IgM, progressive lymphopenia, genetic detection
Mycobacterial Susceptibility (Autosomal Dominant and Recessive Forms)		
Severe extrapulmonary or disseminated infections with bacille Calmette-Guérin (BCG), nontuberculous mycobacteria, salmonella, histoplasmosis, coccidioidomycosis, poor granuloma formation	Inability to kill intracellular organisms due to low IFN-γ production or response. Mutations in IFN-γ receptors, IL-12 receptors, IL-12 p40, *STAT1, NEMO, ISG15, and others*	Abnormally low or very high levels of IFN-γ receptor 1; functional assays of cytokine production and response; genetic detection
GATA2 Deficiency (Autosomal Dominant)		
Persistent or disseminated warts, disseminated mycobacterial disease, low monocytes, NK cells, B cells; hypoplastic myelodysplasia, leukemia, cytogenetic abnormalities, pulmonary alveolar proteinosis	Impaired macrophage activity, cytopenias. Mutations in *GATA2*	Profound circulating monocytopenia, NK and B-cell cytopenias; genetic detection

Abbreviations: C/EBPε, CCAAT/enhancer binding protein-ε; DHR, dihydrorhodamine (oxidation test); DOCK8, dedicator of cytokinesis 8; GI, gastrointestinal; GU, genitourinary; HPV, human papillomavirus; HSV, herpes simplex virus; IFN, interferon; IL, interleukin; IRAK4, IL-1 receptor–associated kinase 4; LFA-1, leukocyte function–associated antigen 1; MyD88, myeloid differentiation primary response gene 88; NADPH, nicotinamide–adenine dinucleotide phosphate; NBT, nitroblue tetrazolium (dye test); NEMO, NF-κB essential modulator; NF-κB, nuclear factor-κB; NK, natural killer; STAT1–3, signal transducer and activator of transcription 1–3; TLR, Toll-like receptor; TNF, tumor necrosis factor.

which induces T-cell production of IL-17, a potent proinflammatory cytokine. These processes conspire to drive inflammation in LAD 1. Patients with LAD 1 have recurrent bacterial infections involving the skin, oral and genital mucosa, and respiratory and intestinal tracts; persistent leukocytosis (resting neutrophil counts of 15,000–20,000/μL) because cells do not marginate; and, in severe cases, a history of delayed separation of the umbilical stump. Infections, especially of the skin, may become necrotic with progressively enlarging borders, slow healing, and development of dysplastic scars. The most common bacteria are *Staphylococcus aureus* and enteric gram-negative bacteria. LAD 2 is caused by an abnormality of fucosylation of SLe^x (CD15s), the ligand on neutrophils that interacts with selectins on endothelial cells and is responsible for neutrophil rolling along the endothelium. Infection susceptibility in LAD 2 appears to be less severe than in LAD 1. LAD 2 is also known as *congenital disorder of glycosylation IIc* (CDGIIc) due to mutation in a GDP-fucose transporter (*SLC35C1*). LAD 3 is characterized by infection susceptibility, leukocytosis, and petechial hemorrhage due to impaired integrin activation caused by mutations in *FERMT3*.

DISORDERS OF NEUTROPHIL GRANULES The most common neutrophil defect is myeloperoxidase deficiency, a primary granule defect inherited as an autosomal recessive trait; the incidence is ~1 in 2000 persons. Myeloperoxidase deficiency is not associated with clinically compromised defenses unless there is another contributing factor such as diabetes. In myeloperoxidase deficiency, neutrophil microbicidal activity is delayed but not absent. Patients with myeloperoxidase deficiency and diabetes are more susceptible to *Candida* infections. An acquired form of myeloperoxidase deficiency occurs in myelomonocytic leukemia and acute myeloid leukemia. Importantly, myeloperoxidase deficiency gives an abnormal dihydrorhodamine (DHR) assay (see chronic granulomatous disease below).

Chédiak-Higashi syndrome (CHS) is a rare disease with autosomal recessive inheritance due to defects in the lysosomal transport protein LYST, encoded by the gene *CHS1* at 1q42. This protein is required for normal packaging and disbursement of granules. Neutrophils (and all cells containing lysosomes) from patients with CHS characteristically have large granules **(Fig. 67-10)**, making it a systemic disease. Patients with CHS have nystagmus, partial oculocutaneous albinism, and an increased number of infections resulting from many bacterial agents. Some CHS patients develop an "accelerated phase" in childhood with a hemophagocytic syndrome and an aggressive lymphoma requiring bone marrow transplantation. CHS neutrophils and monocytes have impaired chemotaxis and abnormal rates of microbial killing due to slow rates of fusion of the lysosomal granules with phagosomes. NK cell function is also impaired. CHS patients may develop a severe disabling peripheral neuropathy in adulthood.

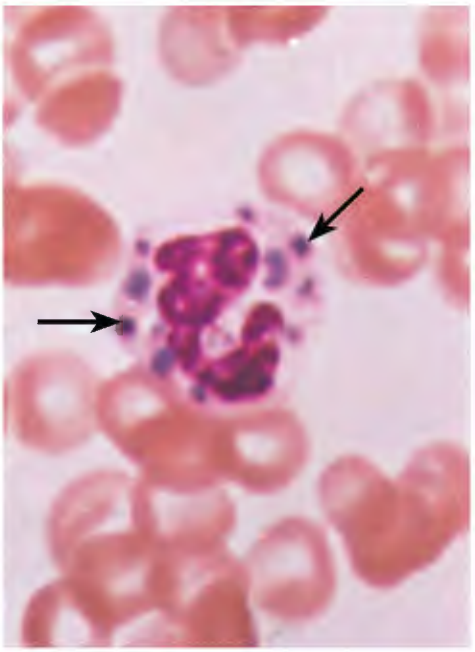


FIGURE 67-10 Chédiak-Higashi syndrome. The granulocytes contain huge cytoplasmic granules formed from aggregation and fusion of azurophilic and specific granules. Large abnormal granules are found in other granule-containing cells throughout the body.

Specific granule deficiency is a rare autosomal recessive disease in which the production of secondary granules and their contents, as well as the primary granule component defensins, is defective. The defect in killing leads to severe bacterial infections. One type of specific granule deficiency is due to a mutation in *CEBPE*, which regulates expression of granule components. A dominant mutation in *CEBPE* has also been described. Specific granule deficiency can also be caused by mutations in *SMARCD2*.

CHRONIC GRANULOMATOUS DISEASE Chronic granulomatous disease (CGD) is a group of genetic disorders of granulocyte and monocyte oxidative metabolism due to defects in the enzyme NADPH oxidase, also called NOX2. CGD has an incidence of ~1 in 100,000–200,000 individuals. In about two-thirds of patients, CGD is inherited as an X-linked recessive trait; the remainder inherit their disease in autosomal recessive patterns. Mutations in the genes encoding the six proteins that allow assembly at the plasma membrane of NOX2 account for all patients with CGD. Two proteins (a 91-kDa protein, abnormal in X-linked CGD, and a 22-kDa protein, absent in one form of autosomal recessive CGD) form the heterodimer cytochrome b-558 in the plasma membrane. The protein *e*ssential for *r*eactive *o*xidant *s*ignaling (EROS) is encoded by *CYBC1*, which is required to transport the 91- and 22-kDa proteins to the endoplasmic reticulum. Three other proteins (40, 47, and 67 kDa, abnormal in the other autosomal recessive forms of CGD) are cytoplasmic and interact with the cytochrome after cell activation to form the NADPH oxidase, which is required for phagocyte hydrogen peroxide production. Therefore, leukocytes from patients with CGD have severely diminished hydrogen peroxide production. Patients with CGD characteristically have increased numbers of infections due to a relatively narrow range of microorganisms (in North America, *S. aureus*, *Serratia marsescens*, *Burkholderia cepacia* complex, and *Nocardia* and *Aspergillus* species). Outside of North America, *Salmonella*, tuberculosis, and bacillus Calmette-Guérin (BCG) are important pathogens. When patients with CGD become infected, they often have extensive inflammatory reactions, and suppuration is common despite the administration of appropriate antibiotics. Aphthous ulcers and chronic infection of the nares are often present. Granulomas are frequent and can obstruct the gastrointestinal or genitourinary tracts. The excessive inflammation is due to failure to downregulate inflammation, reflecting a failure to inhibit the synthesis of, degradation of, or response to interleukins or chemoattractants, leading to persistent myeloid reaction. Impaired killing of intracellular microorganisms by macrophages may lead to persistent cell-mediated immune activation and granuloma formation. Autoimmune complications such as immune thrombocytopenic purpura and juvenile idiopathic arthritis are also increased in CGD. In addition, for unexplained reasons, discoid lupus is more common in X-linked carriers. Late complications, including nodular regenerative hyperplasia and portal hypertension, are increasingly recognized in adolescent and adult patients with CGD. Interestingly, patients with CGD have been reported to be protected from atherosclerosis, suggesting an important role for NADPH oxidase (NOX2) in the pathogenesis of this inflammatory disease of arteries.

DISORDERS OF PHAGOCYTE ACTIVATION Phagocytes depend on cell-surface stimulation to induce signals that evoke multiple levels of the inflammatory response, including cytokine synthesis, chemotaxis, and antigen presentation. Mutations affecting the major pathway that signals through NF-κB have been noted in patients with a variety of infection susceptibility syndromes. If the defects are at a very late stage of signal transduction, in the protein critical for NF-κB activation known as the NF-κB essential modulator (NEMO), then affected males develop ectodermal dysplasia and severe immune deficiency with susceptibility to bacteria, fungi, mycobacteria, and viruses. There are several other proteins that interact with NEMO, mutations in which are

autosomal and can be phenocopies of NEMO deficiency. If the defects in NF-κB activation are closer to the cell-surface receptors, in the proteins transducing Toll-like receptor signals, IL-1 receptor–associated kinase 4 (IRAK4), and myeloid differentiation primary response gene 88 (MyD88), then children have a marked susceptibility to pyogenic infections associated with a striking blunting of the febrile and inflammatory responses early in life but develop resistance to infection later.

The small Rho-family GTPase, RAC2, is involved in actin assembly, intracellular signaling, and superoxide production. Given these multiple roles, it is not surprising that different mutations in *RAC2* can have different effects on its functions. Accordingly, different *RAC2* mutations present, ranging from early-onset sepsis (constitutively active mutations), to abscesses with impaired superoxide production (dominant negative mutations), to common variable immunodeficiency (hyperactive mutations).

MONONUCLEAR PHAGOCYTES

The mononuclear phagocyte system is composed of monoblasts, promonocytes, and monocytes, in addition to the structurally diverse tissue macrophages that make up what was previously referred to as the reticuloendothelial system. Monocytes, which leave the circulation by diapedesis more slowly than neutrophils and have a half-life of 12–24 h, migrate into tissues and differentiate into macrophages. Macrophages are long-lived phagocytic cells capable of many of the functions of neutrophils. They are also secretory cells that participate in many immunologic and inflammatory processes distinct from neutrophils.

Many tissue macrophages ("big eaters") arise in the embryonic yolk sac outside of conventional hematopoiesis and then go on to take up residence in tissues. In addition, there are macrophages derived from monocytes, which may have specialized functions suited for specific anatomic locations. Macrophages are particularly abundant in capillary walls of the lung, spleen, liver, and bone marrow, where they function to remove microorganisms and other noxious elements from the blood. Alveolar macrophages, liver Kupffer cells, splenic macrophages, peritoneal macrophages, bone marrow macrophages, lymphatic macrophages, brain microglial cells, and dendritic macrophages all have specialized functions. Macrophage-secreted products include lysozyme, neutral proteases, acid hydrolases, arginase, complement components, enzyme inhibitors (plasmin, α_2-macroglobulin), binding proteins (transferrin, fibronectin, transcobalamin II), nucleosides, and cytokines (TNF-α; IL-1, 8, 12, 18). IL-1 **(Chaps. 20 and 360)** has many functions, including initiating fever in the hypothalamus, mobilizing leukocytes from the bone marrow, and activating lymphocytes and neutrophils. TNF-α is a pyrogen that duplicates many of the actions of IL-1 and plays an important role in the pathogenesis of gram-negative shock **(Chap. 315)**. TNF-α stimulates production of hydrogen peroxide and related toxic oxygen species by macrophages and neutrophils. In addition, TNF-α induces catabolic changes that contribute to the profound wasting (cachexia) associated with many chronic diseases.

Other macrophage-secreted products include reactive oxygen and nitrogen metabolites, bioactive lipids (arachidonic acid metabolites and platelet-activating factors), chemokines, CSFs, and factors stimulating fibroblast and vessel proliferation. Macrophages help regulate the replication of lymphocytes and participate in the killing of tumors, viruses, and certain bacteria (*Mycobacterium tuberculosis* and *Listeria monocytogenes*). Macrophages are key effector cells in the elimination of intracellular microorganisms. Their ability to fuse to form giant cells that coalesce into granulomas in response to some inflammatory stimuli is important in the elimination of intracellular microbes and is under the control of IFN-γ. Nitric oxide induced by IFN-γ may be an important effector against intracellular parasites, including tuberculosis and *Leishmania*.

Macrophages play an important role in the immune response **(Chap. 360)**. They process and present antigen to lymphocytes and secrete cytokines that modulate and direct lymphocyte development and function. Macrophages participate in autoimmune phenomena by removing immune complexes and other substances from the circulation. Polymorphisms in macrophage receptors for immunoglobulin (FcγRII) determine susceptibility to some infections and autoimmune diseases. In wound healing, they dispose of senescent cells, and they also contribute to atheroma development. Macrophage elastase mediates development of emphysema from cigarette smoking.

DISORDERS OF THE MONONUCLEAR PHAGOCYTE SYSTEM

Many disorders of neutrophils extend to mononuclear phagocytes. Monocytosis is associated with tuberculosis, brucellosis, subacute bacterial endocarditis, Rocky Mountain spotted fever, malaria, and visceral leishmaniasis (kala-azar). Monocytosis also occurs with certain malignancies, leukemias, myeloproliferative syndromes, hemolytic anemias, chronic idiopathic neutropenias, and granulomatous diseases such as sarcoidosis, Crohn disease, and some collagen vascular diseases. Patients with LAD1, hyperimmunoglobulin E–recurrent infection (Job's) syndrome, CHS, and CGD all have defects in the mononuclear phagocyte system.

Monocyte cytokine production or response is impaired in some patients with disseminated nontuberculous mycobacterial infection who are not infected with HIV. Genetic defects in the pathways regulated by IFN-γ and IL-12 lead to impaired killing of intracellular bacteria, mycobacteria, salmonellae, and certain viruses **(Fig. 67-11)**. Autoantibodies to IFN-γ, IL-23, and GM-CSF can block critical signals for macrophage intracellular killing and are associated with mycobacterial, fungal, and *Nocardia* infections.

Certain viral infections impair mononuclear phagocyte function. For example, influenza virus infection causes abnormal monocyte chemotaxis. Mononuclear phagocytes can be infected by HIV using CCR5, the chemokine receptor that acts as a co-receptor with CD4 for HIV. T lymphocytes produce IFN-γ, which induces FcR expression and phagocytosis and stimulates hydrogen peroxide production by mononuclear phagocytes and neutrophils. In certain diseases, such as AIDS, IFN-γ production may be deficient, whereas in other diseases, such as T-cell lymphomas, excessive release of IFN-γ may be associated with erythrophagocytosis by splenic macrophages.

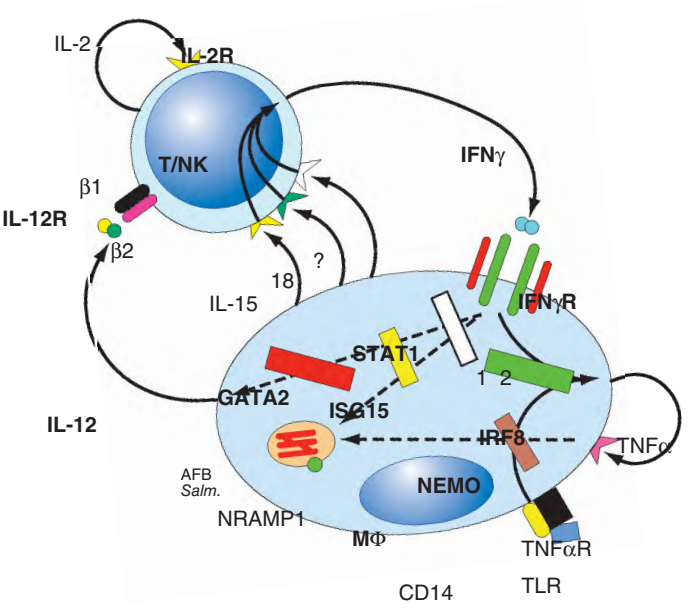


FIGURE 67-11 Lymphocyte-macrophage interactions underlying resistance to mycobacteria and other intracellular pathogens such as *Salmonella, Histoplasma,* and *Coccidioides*. Mycobacteria (and others) infect macrophages, leading to the production of IL-12, which activates T or NK cells through its receptor, leading to production of IL-2 and IFN-γ. IFN-γ acts through its receptor on macrophages to upregulate TNF-γ and IL-12 and kill intracellular pathogens. Other critical interacting molecules include signal transducer and activator of transcription 1 (STAT1), interferon regulatory factor 8 (IRF8), GATA2, and ISG15. Mutant forms of the cytokines and receptors shown in *bold type* have been found in severe cases of nontuberculous mycobacterial infection, salmonellosis, and other intracellular pathogens. AFB, acid-fast bacilli; IFN, interferon; IL, interleukin; NEMO, nuclear factor-κB essential modulator; NK, natural killer; TLR, Toll-like receptor; TNF, tumor necrosis factor.

Autoinflammatory diseases are characterized by abnormal cytokine regulation, leading to excess inflammation in the absence of infection. These diseases can mimic infectious or immunodeficient syndromes. Gain-of-function mutations in the TNF-α receptor cause TNF-α receptor–associated periodic syndrome (TRAPS), which is characterized by recurrent fever in the absence of infection, due to persistent stimulation of the TNF-α receptor **(Chap. 381)**. Diseases with abnormal IL-1 regulation leading to fever include familial Mediterranean fever due to mutations in *PYRIN*. Mutations in *cold-induced autoinflammatory syndrome 1* (*CIAS1*) lead to neonatal-onset multisystem autoinflammatory disease, familial cold urticaria, and Muckle-Wells syndrome. The syndrome of *p*yoderma gangrenosum, *a*cne, and sterile *p*yogenic *a*rthritis (PAPA syndrome) is caused by mutations in *PSTPIP1*. In contrast to these syndromes of overexpression of proinflammatory cytokines, blockade of TNF-α by the antagonists infliximab, adalimumab, certolizumab, golimumab, or etanercept has been associated with severe infections due to tuberculosis, nontuberculous mycobacteria, and fungi **(Chap. 381)**.

Monocytopenia occurs with acute infections, with stress, and after treatment with glucocorticoids. Drugs that suppress neutrophil production in the bone marrow can cause monocytopenia. Persistent severe circulating monocytopenia is seen in GATA2 deficiency, even though macrophages are found at the sites of inflammation. Monocytopenia also occurs in aplastic anemia, hairy cell leukemia, acute myeloid leukemia, and as a direct result of myelotoxic drugs.

EOSINOPHILS

Eosinophils and neutrophils share similar morphology, many lysosomal constituents, phagocytic capacity, and oxidative metabolism. Eosinophils express a specific chemoattractant receptor and respond to a specific chemokine, eotaxin, but little is known about their required role. Eosinophils are much longer lived than neutrophils, and unlike neutrophils, tissue eosinophils can recirculate. During most infections, eosinophils appear unimportant. However, in invasive helminthic infections, such as hookworm, schistosomiasis, strongyloidiasis, toxocariasis, trichinosis, filariasis, echinococcosis, and cysticercosis, the eosinophil plays a central role in host defense. Recently, eosinophils have been recognized as important in the host response to tuberculosis. Eosinophils are associated with bronchial asthma, cutaneous allergic reactions, and other hypersensitivity states.

The distinctive feature of the red-staining (Wright's stain) eosinophil granule is its crystalline core consisting of an arginine-rich protein (major basic protein) with histaminase activity, important in host defense against parasites. Eosinophil granules also contain a unique eosinophil peroxidase that catalyzes the oxidation of many substances by hydrogen peroxide and may facilitate killing of microorganisms.

Eosinophil peroxidase, in the presence of hydrogen peroxide and halide, initiates mast cell secretion in vitro and thereby promotes inflammation. Eosinophils contain cationic proteins, some of which bind to heparin and reduce its anticoagulant activity. Eosinophil-derived neurotoxin and eosinophil cationic protein are ribonucleases that can kill respiratory syncytial virus. Eosinophil cytoplasm contains Charcot-Leyden crystal protein, a hexagonal bipyramidal crystal first observed in a patient with leukemia and then in sputum of patients with asthma; this protein is lysophospholipase and may function to detoxify certain lysophospholipids.

Several factors enhance the eosinophil's function in host defense. T cell–derived factors enhance the ability of eosinophils to kill parasites. Mast cell–derived eosinophil chemotactic factor of anaphylaxis (ECFa) increases the number of eosinophil complement receptors and enhances eosinophil killing of parasites. Eosinophil CSFs (e.g., IL-5) produced by macrophages increase eosinophil production in the bone marrow and activate eosinophils to kill parasites.

EOSINOPHILIA

Eosinophilia is the presence of >500 eosinophils per μL of blood and is common in many settings besides parasite infection. Significant tissue eosinophilia can occur without an elevated blood count. A common cause of eosinophilia is allergic reaction to drugs (iodides, aspirin, sulfonamides, nitrofurantoin, penicillins, and cephalosporins). Allergies such as hay fever, asthma, eczema, serum sickness, allergic vasculitis, and pemphigus are associated with eosinophilia. Eosinophilia also occurs in collagen vascular diseases (e.g., rheumatoid arthritis, eosinophilic fasciitis, allergic angiitis, and periarteritis nodosa) and malignancies (e.g., Hodgkin disease; mycosis fungoides; chronic myeloid leukemia; and cancer of the lung, stomach, pancreas, ovary, or uterus), as well as in dominant negative STAT3 (Job's syndrome), DOCK8 deficiency (see below), and CGD. Eosinophilia is commonly present in helminthic infections. IL-5 is the dominant eosinophil growth factor. Therapeutic administration of the cytokines IL-2 or GM-CSF frequently leads to transient eosinophilia. The most dramatic hypereosinophilic syndromes are Loeffler's syndrome, tropical pulmonary eosinophilia, Loeffler's endocarditis, eosinophilic leukemia, and idiopathic hypereosinophilic syndrome (50,000–100,000/μL). IL-5 is the dominant eosinophil growth factor and can be specifically inhibited with monoclonal antibodies against it or its receptor.

The idiopathic hypereosinophilic syndromes are a heterogeneous group of disorders with the common feature of prolonged eosinophilia of unknown cause and organ system dysfunction, including the heart, central nervous system, kidneys, lungs, gastrointestinal tract, and skin. The bone marrow is involved in all affected individuals, but the most severe complications involve the heart and central nervous system. Clinical manifestations and organ dysfunction are highly variable. Eosinophils are found in the involved tissues and likely cause tissue damage by local deposition of toxic eosinophil proteins such as eosinophil cationic protein and major basic protein. In the heart, the pathologic changes lead to thrombosis, endocardial fibrosis, and restrictive endomyocardiopathy. The damage to tissues in other organ systems is similar. Some cases are due to mutations involving the platelet-derived growth factor receptor, and these are extremely sensitive to the tyrosine kinase inhibitor imatinib. Glucocorticoids, hydroxyurea, and IFN-α each have been used successfully, as have therapeutic antibodies against IL-5 or its receptor. Cardiovascular complications are managed aggressively.

The *eosinophilia-myalgia syndrome* is a multisystem disease, with prominent cutaneous, hematologic, and visceral manifestations, that frequently evolves into a chronic course and can occasionally be fatal. The syndrome is characterized by eosinophilia (eosinophil count >1000/μL) and generalized disabling myalgias without other recognized causes. Eosinophilic fasciitis, pneumonitis, and myocarditis; neuropathy culminating in respiratory failure; and encephalopathy may occur. The disease is thought to be caused by ingesting contaminants in L-tryptophan–containing products. Eosinophils, lymphocytes, macrophages, and fibroblasts accumulate in the affected tissues, but their role in pathogenesis is unclear. Activation of eosinophils and fibroblasts and the deposition of eosinophil-derived toxic proteins in affected tissues may contribute. IL-5 and transforming growth factor β have been implicated as potential mediators. Treatment is withdrawal of products containing L-tryptophan and the administration of glucocorticoids. Most patients recover fully, remain stable, or show slow recovery, but the disease can be fatal in up to 5% of patients.

Eosinophilic neoplasms are discussed in Chap. 115.

EOSINOPENIA

Eosinopenia occurs with stress, such as acute bacterial infection, and after treatment with glucocorticoids. The mechanism of eosinopenia of acute bacterial infection is unknown but is independent of endogenous glucocorticoids, because it occurs in animals after total adrenalectomy. There is no known adverse effect of eosinopenia.

HYPERIMMUNOGLOBULIN E–RECURRENT INFECTION SYNDROME

The hyperimmunoglobulin E–recurrent infection syndrome, Job's syndrome, is a rare multisystem disease in which the immune and somatic systems are affected, including neutrophils, monocytes, T cells, B cells, and osteoclasts. Autosomal dominant negative

mutations in signal transducer and activator of transcription 3 (STAT3) impair normal STAT signaling with broad and profound effects. Patients have characteristic facies with broad nose, kyphoscoliosis, and eczema. Primary teeth erupt normally but do not deciduate, often requiring extraction. Recurrent sinopulmonary and cutaneous infections tend to elicit much less inflammation than appropriate for the degree of infection, leading to "cold abscesses." Pneumonias typically cavitate, leading to pneumatoceles. Coronary artery aneurysms are common, as are cerebral demyelinated plaques that accumulate with age. IL-17–producing T cells, which are thought responsible for protection against extracellular and mucosal infections, are reduced in Job's syndrome. Despite very high IgE levels, Job's patients have only mildly elevated levels of allergy. Patients with autosomal recessive defects in dedicator of cytokinesis 8 (DOCK8) also have very high IgE levels joined to severe allergy, extensive viral susceptibility, and increased rates of cancer. Autosomal dominant *gain-of-function* (GOF) mutations in STAT3 lead to a disease characterized by onset in childhood of lymphadenopathy, autoimmune cytopenias, multiorgan autoimmunity, infections, and interstitial lung disease.

LABORATORY DIAGNOSIS AND MANAGEMENT

A complete blood count (CBC) and differential are essential. Careful examination of neutrophils on peripheral blood smears can diagnose CHS and suggest other neutrophil granule abnormalities such as specific granule deficiency. Bone marrow examination, where indicated, should be accompanied by either gene panel or whole exome/genome sequencing if genetic defects are suspected. Bone marrow reserves (steroid challenge test), marginated circulating pool of cells (epinephrine challenge test), and marginating ability (endotoxin challenge test) are also doable (Fig. 67-8). In vivo assessment of inflammation is possible with a Rebuck skin window test or an in vivo skin blister assay, which measure the ability of leukocytes and inflammatory mediators to accumulate locally in the skin. In vitro tests of phagocyte aggregation, adherence, chemotaxis, phagocytosis, degranulation, and microbicidal activity may help pinpoint cellular or humoral lesions. CGD is detected with either the nitroblue tetrazolium (NBT) dye test or the DHR oxidation test. These tests are based on the ability of products of oxidative metabolism to alter the oxidation states of reporter molecules so that they can be detected microscopically (NBT) or by flow cytometry (DHR). While the DHR is very sensitive for CGD, impaired DHR responses are seen in myeloperoxidase deficiency and with acetaminophen ingestion.

Patients with leukopenias or leukocyte dysfunction often have delayed inflammatory responses. Therefore, clinical manifestations may be minimal despite overwhelming infection, and unusual infections must always be suspected. Early signs of infection demand prompt, aggressive culturing for microorganisms, use of antibiotics, and drainage of abscesses. Prolonged courses of antibiotics are often required. In patients with CGD, prophylactic antibiotics (trimethoprim-sulfamethoxazole) and antifungals (itraconazole) markedly diminish the frequency of life-threatening infections. Glucocorticoids may relieve gastrointestinal or genitourinary tract obstruction by granulomas in patients with CGD. Although TNF-α-blocking agents may markedly relieve inflammatory bowel symptoms, extreme caution must be exercised in their use in CGD inflammatory bowel disease, because it profoundly increases these patients' already heightened susceptibility to infection. Recombinant human IFN-γ, which nonspecifically stimulates phagocytic cell function, reduces the frequency of infections in patients with CGD by 70% and reduces the severity of infection. This effect of IFN-γ in CGD is additive to the effect of prophylactic antibiotics. The recommended dose is 50 $\mu g/m^2$ subcutaneously three times weekly. IFN-γ has also been used successfully in the treatment of leprosy, nontuberculous mycobacteria, and visceral leishmaniasis.

Rigorous oral hygiene reduces but does not eliminate the discomfort of gingivitis, periodontal disease, and aphthous ulcers; chlorhexidine mouthwash and tooth brushing with a hydrogen peroxide–sodium bicarbonate paste also helps many patients. Oral antifungal agents (fluconazole, itraconazole, voriconazole, posaconazole) have reduced mucocutaneous candidiasis in patients with Job's syndrome. Recombinant G-CSF is useful in the management of certain forms of neutropenia due to depressed neutrophil production, including those related to cancer chemotherapy. Patients with chronic neutropenia with evidence of a good bone marrow reserve need not receive prophylactic antibiotics. Patients with chronic or cyclic neutrophil counts <500/μL may benefit from prophylactic antibiotics and G-CSF during periods of neutropenia. Oral trimethoprim-sulfamethoxazole (160/800 mg) twice daily can prevent infection. Increased numbers of fungal infections are not seen in patients with CGD on this regimen. Oral quinolones such as levofloxacin and ciprofloxacin are alternatives.

In the setting of cytotoxic chemotherapy with severe, persistent lymphocyte dysfunction, trimethoprim-sulfamethoxazole prevents *Pneumocystis jirovecii* pneumonia. These patients, and patients with phagocytic cell dysfunction, should avoid heavy exposure to airborne soil, dust, or decaying matter (mulch, manure), which are often rich in *Nocardia* and the spores of *Aspergillus* and other fungi. Restriction of activities or social contact has no proven role in reducing risk of infection for phagocyte defects.

Although aggressive medical care for many patients with phagocytic disorders can allow them to go for years without a life-threatening infection, there may still be delayed effects of prolonged antimicrobials and other inflammatory complications. Cure of most congenital phagocyte defects is possible by bone marrow transplantation, and rates of success are improving **(Chap. 119)**. The identification of specific gene defects in patients with LAD1, CGD, and other immunodeficiencies has led to gene therapy trials in a number of genetic white cell disorders.

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68 Primary and Secondary Hemophagocytic Lymphohistiocytosis

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Hemophagocytic lymphohistiocytosis (HLH) is a highly life-threatening condition of massive, uncontrolled inflammation (a hyperinflammation). It exists in a primary (genetic) form and a secondary (acquired) form, and is often triggered by infections, malignancies, and autoimmune diseases. HLH triggered by autoimmune conditions is also referred to as macrophage activation syndrome (MAS or MAS-HLH). While the underlying factors are biologically very different, the clinical picture of all forms of HLH is quite similar, with a sepsis-like inflammation often leading to multiple organ failure and death if not treated promptly and appropriately (convergent phenotype). Primary HLH mostly affects children, whereas secondary HLH is far more common in adults. The current chapter will present primary and secondary HLH separately. A classification of HLH is presented in Table 68-1.

TABLE 68-1 Simplified Classification of Hemophagocytic Lymphohistiocytosis (HLH)

Primary HLH: Mendelian Conditions
Lymphocyte cytotoxic defects
FHL2 (*PRF1*)
FHL3 (*UNC13D*)
FHL4 (*STX11*)
FHL5 (*STXBP2*)
XLP1 (*SH2D1A*)
Griscelli syndrome type 2 (*RAB27A*)
Chédiak-Higashi syndrome (*LYST*)
Abnormalities of inflammasome activation
XLP2 (*BIRC4*)
NLRC4
Mendelian disorders affecting inflammation
Lysinuric protein intolerance (*SLC7A7*)
Wolman disease (*LIPA*)
Secondary HLH (Non-Mendelian)
Infection-associated HLH
Virus-associated HLH
Bacteria-associated HLH
Parasite-associated HLH
Fungal-associated HLH
Malignancy-associated HLH
Malignancy-triggered HLH
HLH occurring during chemotherapy
Autoimmune-associated HLH (MAS-HLH)
HLH with SoJIA
HLH with adult-onset Still's disease
HLH with SLE
HLH with vasculitis
Transplant-associated HLH

Abbreviations: MAS, macrophage activation syndrome; SLE, systemic lupus erythematosus; SoJIA, systemic-onset juvenile idiopathic arthritis.

Source: Reproduced with permission from JF Emile et al: Revised classification of histiocytoses and neoplasms of the macrophage-dendritic cell lineages. Blood 127: 2672, 2016.

PRIMARY HEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS

Primary HLH comprises Mendelian-inherited conditions causing HLH. The most common form, by far, is familial HLH (FHL), which is associated with lymphocyte cytotoxicity defects.

Other forms of primary HLH associated with lymphocyte cytotoxicity defects are Griscelli syndrome type 2 (associated with hypopigmentation, silver-colored hair, and immune defects due to mutation in *RAB27A*, a *RAS* gene family member), Chédiak-Higashi syndrome, and X-linked lymphoproliferative syndrome (XLP) type 1. Another group of primary HLHs are associated with abnormalities of inflammasome activation, inborn errors of immunity, and inborn errors of metabolism.

FAMILIAL HEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS

If not treated, FHL is rapidly fatal, with a median survival of 1–2 months. The incidence of FHL has been estimated to be 1 in 50,000 live births. Together with severe combined immunodeficiency, it is the most common fatal inherited human immunodeficiency. Because its inheritance is autosomal recessive, it is more common in regions where consanguinity is frequent; it is equally common in males and females. The median age at onset is between 3 and 6 months. It may be diagnosed at birth or even in utero, whereas less severe forms may result in onset during adolescence and adulthood.

Clinical and Laboratory Features A typical presentation is a fever, often in an infant, with a sepsis-like condition associated with cytopenias, in particular thrombocytopenia, and hepatosplenomegaly. If blood cultures are negative and signs of malignancy are absent, HLH should be considered and assessed for promptly.

HEMATOLOGIC AND BONE MARROW FINDINGS Thrombocytopenia is frequent, and the platelet levels may be extremely low. Anemia and neutropenia are also common, and bicytopenia (i.e., at least two of these three lineages affected) is a diagnostic criterion.

Hemophagocytosis is not always present, particularly in the early phase of the disease, and hence, serial examinations over time may be helpful. However, hemophagocytosis is not specific for HLH. Over time, the bone marrow may become hypocellular.

LIVER INVOLVEMENT AND COAGULOPATHY Ferritin and transaminase levels are almost always elevated. Ferritin, an acute-phase reactant (i.e., a marker of inflammation), can serve as a rapid and easily available screening test for HLH, although it is not specific for the disease. Icterus, with a predominantly conjugated hyperbilirubinemia, and elevated γ-glutamyl transferase levels are also common. HLH may resemble acute liver failure, and in histopathologic specimens, an image similar to chronic persistent hepatitis may be found. Disseminated intravascular coagulation, severe acute bleeding, and low levels of fibrinogen are frequent.

CENTRAL NERVOUS SYSTEM INVOLVEMENT About one a third of children with FHL have neurologic symptoms at diagnosis, which may be severe, and around half have abnormal cerebrospinal fluid with elevated cell and/or protein content. Common symptoms include seizures, decreased consciousness, and signs of meningismus, as well as irritability and hypotonia in neonates and infants. Cranial nerve palsies and ataxia may also occur. Isolated neurologic symptoms may be the first presentation of FHL, particularly in older children and adolescents. Magnetic resonance imaging findings are nonspecific.

OTHER CLINICAL AND LABORATORY FINDINGS Petechia, purpura, and a nonspecific transient maculopapular rash are frequent. Lymph node enlargement develops in about half of the children. Genetic defects in *STXBP2* may cause gastrointestinal symptoms with severe diarrhea that persists despite successful hematopoietic stem cell transplantation (HSCT). Triglyceride levels are often significantly elevated, and sodium and albumin levels decreased. Soluble interleukin (IL) 2 receptor (sIL-2r; also known as sCD25) levels are often markedly